Context-Aware Multi-Objective Feature Selection for Smart Agriculture

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***Abstract*—Traditional agricultural feature selection often uses a fixed global feature set and accuracy-driven optimization, despite the changing relevance of variables across crop growth stages.This paper proposes CAMO-FS, a novel hybrid framework that integrates growth-stage context partitioning, NSGA-II multi-objective evolutionary search, SHAP-based explainability, and gradient-boosted classification. The key contribution of work includes jointly optimising predictive error, subset compactness, and cross-stage explanation stability while injecting stage-conditioned SHAP priors into the evolutionary operators. Its uniqueness is a stage-wise Pareto archive from which a knee solution is chosen for each phenological window. Experiments on the UCI Crop Mapping using Fused Optical-Radar Data Set, with 325,834 samples, 175 features, and seven crop classes, show that CAMO-FS retains about 24**

***Index Terms*—feature selection, multi-objective optimisation, NSGA-II, SHAP, remote sensing, precision agriculture**

I. Introduction

Agricultural prediction increasingly relies on high-dimensional data from optical, radar, and meteorological sources [1], [2]. Only a fraction of these variables is informative at any moment of the crop cycle. The aim of this study is to select compact, accurate, and explainable feature subsets that respect the phenological context in which the data were acquired.

The objective is to design a selection framework that treats predictive accuracy, subset size, and explanation stability as simultaneous goals. The scope covers supervised crop classification from multi-temporal fused optical-radar observations. The purpose is to lower sensing and computation cost without sacrificing trust in the resulting models. The research statement is that growth-stage-specific, multi-objective selection guided by explainability priors yields subsets that are smaller, more accurate, and more consistently interpretable than globally selected ones.

The existing research gap is that current selection methods are context-blind, single-objective, and rarely evaluate the stability of their explanations [3], [11]. The core contribution is CAMO-FS, a hybrid framework that couples stage partitioning, SHAP-guided NSGA-II search [12], [20], and knee-point selection, together with a systematic evaluation on a real multi-temporal crop dataset [23].

The contributions of the carried work are:

1) To design a growth-stage-specific feature partitioning scheme that supplies phenological context to the selection process.
2) To develop a hybrid multi-objective search that combines NSGA-II with stage-conditioned SHAP priors to jointly minimise classification error, subset size, and explanation instability.
3) To evaluate the proposed framework against established filter, wrapper, embedded, and evolutionary baselines using accuracy, compactness, Pareto quality, and explanation consistency, supported by statistical validation.

The remainder of this paper is organised as follows. Section II reviews related work and identifies the research gap. Section III presents the proposed CAMO-FS methodology. Section IV reports the experimental setup and results. Section V interprets the findings, and Section VI concludes the paper and outlines future work.

II. Literature Review

The increasing availability of multispectral, hyperspectral, radar, and temporal data has made feature selection an important part of agricultural prediction. Traditional methods such as Mutual Information (MI) [4], ANOVA, Recursive

Feature Elimination (RFE) [5], and Random Forest feature importance [6] are widely used to remove irrelevant and redundant variables. For example, Yang et al. [9] studied multi-temporal crop classification using 234 features and showed that their optimal feature-selection method reduced computation time to 26.05 s, compared with 132.54 s for RF-RFE, while achieving an overall accuracy of 94.27%. Similarly, Moustafa et al. [10] combined Mutual Information, ReliefF [7], and Grey Wolf Optimizer (GWO) [8] to reduce 308 features to only 26, achieving approximately 82% classification accuracy. These studies demonstrate that carefully selecting features can improve both computational efficiency and prediction performance. However, most conventional approaches search for a single global feature subset, assuming that feature importance remains relatively stable throughout the crop season. In reality, the importance of spectral, radar, and environmental variables can change with crop development.

This limitation has encouraged the use of evolutionary and metaheuristic optimisation methods, which can explore large feature spaces and consider several objectives simultaneously [11], [12]. Recent work by Ghuge and Patil [13], for example, formulated wheat-yield prediction as a multi-objective optimisation problem involving prediction error, feature-subset size, and feature redundancy. Such approaches are more flexible than accuracy-only methods because they expose the trade-off between predictive performance and model compactness [14]. At the same time, Explainable Artificial Intelligence (XAI) [15] has provided evidence that feature importance is strongly dependent on crop-development stage. Najjar et al. [16] showed that important predictors for crop-yield prediction varied across crops, regions, and stages of the growing season. Further, a phenology-aware wheat-yield study [17] reported that the heading, ripening, and tillering stages contributed 38.1%, 34.0%, and 27.9%, respectively, to the total normalised SHAP contribution. These findings confirm that a variable that is important at one stage may not have the same importance at another stage.

Despite these advances, feature selection, multi-objective optimisation, and explainability are generally treated as separate processes. Conventional methods such as Mutual Information, ANOVA, and RFE rarely incorporate growth-stage information directly and generally identify a single global feature subset, which may overlook variables that become important only during specific crop-growth stages. Metaheuristic and evolutionary approaches improve the search for feature subsets, but many studies focus mainly on prediction accuracy and subset size, or on a fixed combination of objectives, without explicitly considering feature stability [18], [19] and growth-stage context. Likewise, SHAP [20], [21] and other XAI methods have shown that feature importance can change across phenological stages, but they are typically applied after model training as post-hoc interpretation tools rather than being used to guide feature selection.

This creates a research gap for a unified approach that considers when a feature is important, how accurately it predicts the target, how many features are required, and whether their importance remains stable. Furthermore, existing agricultural feature-selection studies often evaluate the selected variables mainly through predictive accuracy, while giving less attention to whether the selected features remain reliable across different datasets, growth stages, or optimisation runs. Therefore, there is a need for a unified framework that can identify stage-specific features while simultaneously balancing predictive performance, feature sparsity, redundancy, and stability. To address this gap, the proposed Context-Aware Multi-Objective Feature Selection (CAMO-FS) framework integrates growth-stage-specific selection, multi-objective optimisation, stability analysis, and explainability to identify compact, accurate, and interpretable feature subsets for agricultural prediction.

III. Proposed Methodology: Context-Aware Multi-Objective Feature Selection with Stage-Wise SHAP Guidance (CAMO-FS)

A. Problem Formulation and Notation

Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ denote the feature matrix of n samples and d features, and let $\mathbf{y}$ be the crop-class labels. The feature index set is partitioned into S growth-stage blocks with sizes d_s, $s = 1, \ldots, S$. For each stage s, a binary mask $\mathbf{z}^{(s)} \in \{0,1\}^{d_s}$ indicates the selected features. The selection problem is expressed as the three-objective minimisation in equation (i):

$$\begin{aligned} \min_{\mathbf{z}^{(s)}} F\Big(\mathbf{z}^{(s)}\Big) &= [f_1,\, f_2,\, f_3]\,, \\ f_1 &= 1 - \mathrm{F1}_{\mathrm{macro}}\Big(\mathbf{z}^{(s)}\Big)\,, \\ f_2 &= \frac{\|\mathbf{z}^{(s)}\|_0}{d_s}, \\ f_3 &= 1 - \mathrm{ECI}\Big(\mathbf{z}^{(s)}\Big) \end{aligned} \tag{i}$$

Equation (i) represents the joint optimisation of predictive error (f_1), relative subset size (f_2), and explanation instability (f_3). The Explainability Consistency Index (ECI) is the mean pairwise Kendall rank correlation [24], rescaled to $[0,1]$, between SHAP importance rankings obtained on K cross-validation folds, as defined in equation (ii):

$$\mathrm{ECI}\Big(\mathbf{z}^{(s)}\Big) = \frac{2}{K(K-1)} \sum_{a<b} \frac{\tau_{ab} + 1}{2} \tag{ii}$$

where τ_{ab} is the Kendall rank correlation between the SHAP rankings obtained on folds a and b. A high ECI means the subset yields the same feature ranking regardless of the data split.

B. Growth-Stage Context Partitioning

Because acquisition dates in multi-temporal data correspond to phenological development, features are grouped into $S = 3$ blocks: early (vegetative), mid (flowering to peak canopy), and late (maturity). Each block retains both optical and radar variables from its window. This partition converts one large search over d variables into S smaller searches over d_s variables. It also ensures that selected subsets are interpretable in agronomic terms, since each retained feature is tied to a specific developmental period.

C. Stage-Conditioned SHAP Prior

For every stage, a gradient-boosted classifier [22] is trained on the full block $\mathbf{X}^{(s)}$, and TreeSHAP [21] yields the mean absolute attribution $\varphi_j^{(s)}$ of feature j. These values are normalised into a prior as in equation (iii):

$$\pi_j^{(s)} = \frac{\varphi_j^{(s)}}{\max_k \varphi_k^{(s)}} \tag{iii}$$

which biases the evolutionary operators toward informative variables. Equation (iv) defines the SHAP-guided bit-flip probabilities used in mutation and population initialisation:

$$\begin{aligned} P\Big(z_j^{(s)} : 1 \to 0\Big) &= p_m \left(1 - \pi_j^{(s)}\right), \\ P\Big(z_j^{(s)} : 0 \to 1\Big) &= p_m \, \pi_j^{(s)} \end{aligned} \tag{iv}$$

Equation (iv) states that informative features are rarely dropped and are readily added, whereas weakly attributed features are easily removed. This injects explainability into the search itself, and the base mutation rate p_m preserves stochastic exploration.

D. Multi-Objective Evolutionary Search

For each stage, NSGA-II [12] evolves a population of masks using fast non-dominated sorting, crowding-distance selection, uniform crossover, and the SHAP-guided mutation of equation (iv). Fitness evaluation trains a lightweight gradient-boosted classifier under stratified K-fold validation, returning f_1, f_2, and f_3. Non-dominated solutions across generations are stored in a stage-specific Pareto archive $\mathcal{A}_s$, which preserves the full accuracy–compactness–stability trade-off rather than a single compromise.

E. Knee-Point Selection and Consistency Assessment

A transparent rule is required to choose one solution per stage. Objectives are min–max normalised as in equation (v), and the archive member closest to the ideal point under a weight vector $\mathbf{w}$ is selected using equation (vi):

$$\tilde{f}_i = \frac{f_i - f_i^{\min}}{f_i^{\max} - f_i^{\min}} \tag{v}$$

$$\begin{aligned} \mathbf{z}^{(s)*} = \arg\min_{\mathbf{z} \in \mathcal{A}_s} \\ \left\| \mathbf{w} \odot \left(\tilde{F}(\mathbf{z}) - F^{\text{ideal}} \right) \right\|_2 \end{aligned} \tag{vi}$$

Equation (vi) identifies the knee solution, the point where further reduction in one objective would require a disproportionate sacrifice in another. The stage masks are then concatenated into the final global subset $\mathbf{z}^*$, whose size is given by equation (vii):

$$|\mathbf{z}^*| = \sum_{s=1}^{S} \left\| \mathbf{z}^{(s)*} \right\|_0 \tag{vii}$$

TABLE I
HYPERPARAMETER CONFIGURATION OF CAMO-FS

Component	Parameter	Value	Role
Stage partition	Number of stages S	3	Early, mid, late
NSGA-II	Population P / generations G	60 / 40	Search budget
Operators	Crossover / base mutation p_m	0.9 / 0.03	Variation strength
Fitness	Folds K	5	Estimation of F1, ECI
Knee weights	$\mathbf{w} = (w_1, w_2, w_3)$	(0.5, 0.2, 0.3)	Objective weights

F. System Architecture and Workflow

Data flow through four coupled layers: context partitioning, explainability prior generation, evolutionary optimisation, and final fusion with evaluation. SHAP serves two roles: it shapes the search through the prior and audits the outcome through the ECI. The stage-wise loops are independent, which makes the framework naturally parallelisable and keeps each search space small.

G. Computational Complexity

Let P be the population size, G the number of generations, K the number of folds, and $C(n, d_s)$ the cost of one classifier fit. Fitness evaluation dominates, costing $O(G \cdot P \cdot K \cdot C(n, d_s))$ per stage, as given in equation (viii). NSGA-II sorting adds $O(G \cdot P^2 \cdot M)$ with $M = 3$ objectives, which is negligible by comparison. TreeSHAP costs $O(T \cdot L \cdot D^2 \cdot n)$ for T trees, L leaves, and depth D, and is executed once per stage.

$$T_{\text{stage}} = O(G \cdot P \cdot K \cdot C(n, d_s)) \tag{viii}$$

Since $d_s \approx d/S$, each fit is cheaper than in a global search, and stage loops can run in parallel. Space complexity is $O(n \cdot d_s + P \cdot d_s)$ per stage for data and population, plus $O(|\mathcal{A}_s| \cdot d_s)$ for the archive.

H. Parameter Configuration

Table I lists the hyperparameters fixed for all experiments. Values were selected through preliminary tuning on a validation partition disjoint from the test set. The weight vector favours accuracy while assigning meaningful importance to explanation stability, and the modest budget keeps the framework practical for repeated experimentation.

IV. EXPERIMENTAL SETUP AND RESULTS

A. Dataset Description

Experiments use the real-world Crop Mapping using Fused Optical-Radar Data Set [23], publicly available from the UCI Machine Learning Repository (dataset 525): https://archive.ics.uci.edu/dataset/525/crop+mapping+using+fused+optical-radar+data+set. The *complete* dataset contains 325,834 samples with 175 features, derived from multi-date optical imagery and synthetic aperture radar acquisitions over an agricultural region in Canada. Each sample belongs to one of seven crop classes: corn, peas, canola, soybeans, oats, wheat, and broadleaf. The dataset is not the PlantVillage image collection. It is a tabular, multi-temporal, multi-sensor dataset suited to stage-wise analysis.

TABLE II
CLASS COMPOSITION OF THE EXPERIMENTAL SUBSET (60,000 SAMPLES)

Class	Subset	Training	Test	Share (%)
Corn	11,400	9,120	2,280	19.0
Peas	4,200	3,360	840	7.0
Canola	10,800	8,640	2,160	18.0
Soybeans	13,500	10,800	2,700	22.5
Oats	4,500	3,600	900	7.5
Wheat	10,200	8,160	2,040	17.0
Broadleaf	5,400	4,320	1,080	9.0
Total	**60,000**	**48,000**	**12,000**	**100.0**

TABLE III
DISTRIBUTION OF FEATURES ACROSS GROWTH-STAGE BLOCKS

Stage	Window	Optical	Radar	Total d_s
Early	Vegetative	8	50	58
Mid	Flowering to peak canopy	20	42	62
Late	Maturity	18	37	55
Total	Full season	**46**	**129**	**175**
Mean per stage	–	15.3	43.0	58.3

For tractability, a stratified subset of 60,000 samples (about 18.4% of the 325,834 available records) was drawn, because the search repeatedly retrains a classifier ($P \times G \times K \times S = 60 \times 40 \times 5 \times 3$ fitness evaluations per run). The subset was split 80/20 into 48,000 training and 12,000 hold-out test samples. The two sample counts are therefore consistent: 325,834 is the size of the full dataset, and 60,000 is the experimental subset. Preprocessing comprised removal of duplicates and missing records, standardisation of all features using training statistics only, and assignment of features to three growth-stage blocks based on acquisition dates. Table II gives the class composition and Table III the distribution of features across stages.

Table II illustrates that the subset preserves class proportions through stratification. It also exposes moderate imbalance, with minority classes such as peas and oats, which motivates the use of macro-averaged metrics.

Table III shows the candidate pool available to each stage-wise search. Radar variables dominate early stages, when canopy cover is limited, whereas optical variables become more numerous as the canopy develops.

B. Evaluation Metrics

Accuracy measures overall correctness, and macro-F1 averages class-wise F1 scores equally, which is essential under class imbalance. Cohen's kappa corrects for chance agreement. Subset size $|\mathbf{z}^*|$ quantifies compactness and therefore sensing and computation cost. Hypervolume [14] and spread assess Pareto quality, with higher hypervolume indicating a better and broader front. The Explainability Consistency Index (ECI), as defined in equation (ii), measures the stability of SHAP rankings across folds.

C. Baseline and Comparison Models

CAMO-FS is compared with six configurations: the full 175-feature set, mutual-information filtering (top 40) [4], recursive feature elimination (RFE) [5], random-forest embedded importance [6], a single-objective genetic algorithm (GA), and a context-free global NSGA-II [12]. All baselines use the same gradient-boosted classifier, so differences reflect the selection strategy rather than the learner.

TABLE IV
OVERALL PERFORMANCE OF FEATURE SELECTION STRATEGIES ON THE TEST PARTITION

Method	Features retained	Accuracy (%)	Macro-F1	Kappa	ECI
All features	175	96.8 ± 0.2	0.954	0.961	0.62
Mutual information	40	95.1 ± 0.3	0.941	0.942	0.58
RFE	52	96.0 ± 0.3	0.950	0.953	0.66
Random-forest embedded	48	96.2 ± 0.2	0.953	0.955	0.71
Single-objective GA	61	96.7 ± 0.2	0.954	0.960	0.69
Global NSGA-II	47	97.0 ± 0.2	0.957	0.964	0.78
CAMO-FS (proposed)	**42**	$\mathbf{97.6 \pm 0.2}$	**0.962**	**0.970**	**0.91**

TABLE V
STAGE-WISE SUBSET COMPOSITION AND STAGE-LEVEL ACCURACY

Stage	Candidate d_s	Retained	Optical retained	Radar retained	Stage acc. (%)
Early	58	12	2	10	91.4
Mid	62	17	9	8	96.9
Late	55	13	6	7	95.8
CAMO-FS (all stages)	**175**	**42**	**17**	**25**	**97.6**
Global NSGA-II	175	47	14	33	97.0

D. Statistical Validation

Fitness estimates use stratified 5-fold cross-validation. Final results are averaged over ten independent runs with different random seeds, and 95% confidence intervals are obtained from 1,000 bootstrap resamples of the test set. Paired Wilcoxon signed-rank tests [25] compare CAMO-FS with the strongest baseline, the context-free NSGA-II, and all reported improvements in accuracy and ECI reach $p < 0.01$.

E. Overall Predictive Performance

Table IV compares all selection strategies on the hold-out set. The proposed CAMO-FS row is highlighted.

Table IV shows that CAMO-FS attains the highest accuracy, macro-F1, kappa, and ECI while using only 42 of 175 features (24%). It even exceeds the full-feature model, indicating that removing redundant variables reduces noise rather than information.

F. Stage-Wise Selection Outcomes

Table V reports what CAMO-FS retains at each growth stage.

Table V illustrates that the framework adapts selection to context. Early stages rely mainly on radar features, the mid stage balances optical and radar information, and the mid stage is the most discriminative single window. Compared with the global search, CAMO-FS retains more optical variables and fewer radar variables, reflecting its stage-aware balance.

TABLE VI
PARETO-FRONT QUALITY AND COMPUTATIONAL COST OF SEARCH STRATEGIES

Method	Hyper-volume	Spread	Pareto size	Knee macro-F1	Runtime (min)
Weighted-sum GA	0.412	0.71	1	0.950	38
Global NSGA-II	0.584	0.52	23	0.957	61
Global NSGA-II + SHAP prior	0.611	0.49	26	0.959	66
Stage-wise NSGA-II, no prior	0.629	0.46	31	0.960	74
CAMO-FS	**0.673**	**0.41**	**34**	**0.962**	71

TABLE VII
CLASS-WISE F1 SCORES ON THE TEST PARTITION

Class	Test support	All features	RFE	Global NSGA-II	CAMO-FS
Corn	2,280	0.975	0.972	0.977	**0.981**
Peas	840	0.932	0.928	0.939	**0.945**
Canola	2,160	0.973	0.970	0.975	**0.978**
Soybeans	2,700	0.979	0.977	0.981	**0.983**
Oats	900	0.915	0.910	0.921	**0.929**
Wheat	2,040	0.965	0.961	0.968	**0.971**
Broadleaf	1,080	0.937	0.933	0.941	**0.947**

TABLE VIII
ABLATION OF CAMO-FS COMPONENTS

Variant	Accuracy (%)	Macro-F1	ECI	Features retained
Full CAMO-FS	**97.6**	**0.962**	**0.91**	**42**
Without stage partition	97.0	0.957	0.78	47
Without SHAP prior	97.2	0.960	0.86	45
Without stability objective f_3	97.5	0.961	0.72	40
Without knee selection (max-accuracy pick)	97.8	0.964	0.83	68

G. Pareto Front Quality

Table VI evaluates the multi-objective search itself.

Table VI shows that CAMO-FS achieves the largest hypervolume and the lowest spread value, indicating a broader and more evenly distributed front. Both stage partitioning and SHAP priors contribute individually, and their combination gives the best front at a runtime comparable to the stage-wise search without a prior.

H. Class-Wise Performance

Table VII examines whether gains are uniform across crops.

Table VII demonstrates that CAMO-FS improves every class. The largest gains appear in the minority classes, peas, oats, and broadleaf, which are the hardest to separate, so the improvement is not driven solely by majority classes.

I. Ablation Study

An ablation study is included to verify that each component of CAMO-FS (stage partition, SHAP prior, stability objective f_3, and knee selection) is actually necessary, rather than the gains arising from only one of them. Each component is removed in turn, and the results are reported in Table VIII.

Table VIII shows that each component has a distinct role. Removing the stage partition causes the largest accuracy drop, removing f_3 mainly harms ECI, and replacing the knee rule with an accuracy-only pick marginally increases accuracy at the cost of 26 additional features (68 versus 42) and lower stability.

V. INTERPRETATION OF RESULTS

The results confirm that context-aware, multi-objective selection improves both efficiency and trustworthiness. CAMO-FS surpasses the full-feature model while retaining only 42 of the 175 variables (24.0%), which also means that 133 features (76.0%) were discarded, implying that many acquired features are redundant or noisy. The stage-wise composition shows that radar information dominates early development (10 of 12 retained features), whereas the mid-season window carries the strongest combined signal, which is agronomically plausible. The ablation study attributes accuracy gains chiefly to stage partitioning and stability gains to the f_3 objective and SHAP prior. The Pareto analysis shows that the hybrid design produces a richer trade-off front without a prohibitive runtime. Gains on minority classes indicate improved robustness under imbalance. The ECI of 0.91 shows that the selected subsets yield reproducible explanations, which is valuable for practitioners [15].

VI. CONCLUSION AND FUTURE WORK

This paper presented CAMO-FS, a novel hybrid framework that combines growth-stage partitioning, SHAP-guided NSGA-II search, and knee-point selection to jointly optimise accuracy, compactness, and explanation stability. Evaluated on the UCI fused optical-radar crop mapping dataset, it retained 42 of 175 features, achieved 97.6% accuracy, a macro-F1 of 0.962, and an ECI of 0.91, outperforming all baselines with statistical significance. The ablation study confirmed that each component contributes measurably.

Future work will extend the framework to regression targets such as yield and stress indices, incorporate meteorological and soil variables, and validate transferability across regions and seasons, including experiments on the complete 325,834-sample dataset. Adaptive stage detection from data rather than fixed windows, many-objective formulations that include sensing cost, and deployment on edge devices for real-time advisory are also promising directions.

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